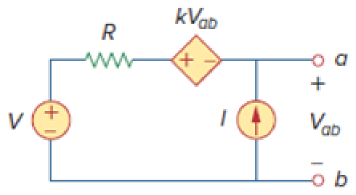
Chapter 4, Problem 10P

Problem

Using Fig. 4.78, design a problem to help other students better understand superposition. Note, the letter *k* is a gain you can specify to make the problem easier to solve but must not be zero.

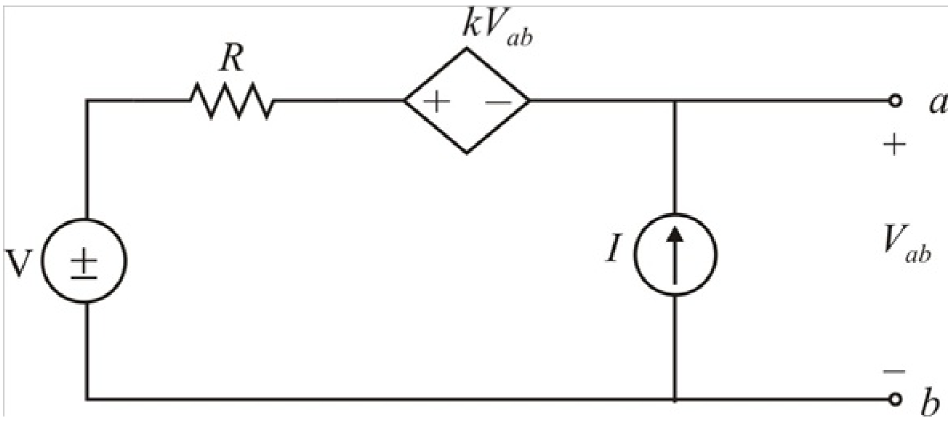
Figure. 4.78:



Step-by-step solution

Step 1 of 5

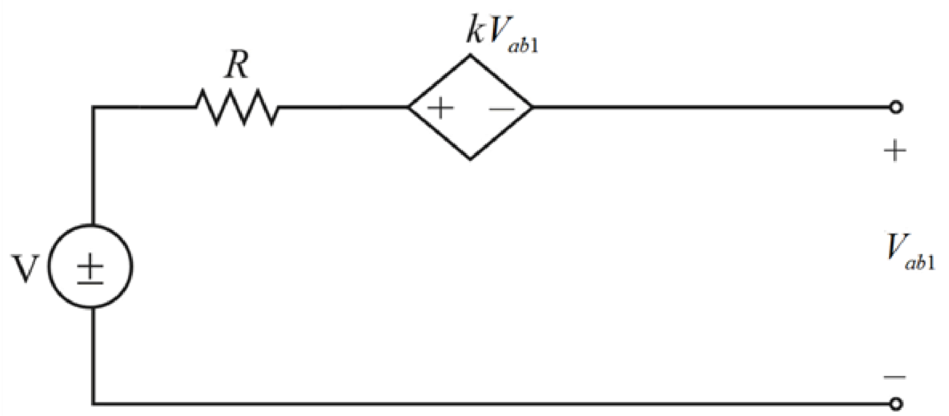
Considering the circuit



Step 2 of 5

From superposition theorem

Let the current source is absent, now the circuit is



Step 3 of 5

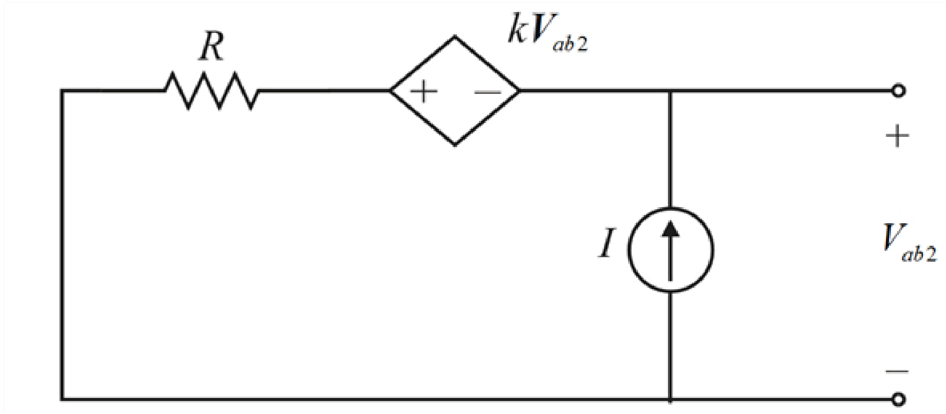
$$-V_{ab1} - kV_{ab1} + V = 0$$

$$-V_{ab1}(1+k) + V = 0$$

$$V_{ab1} = \frac{V}{1+k}$$

Step 4 of 5

Let the voltage source is absent, now the circuit is



Step 5 of 5

$$-V_{ab2} - kV_{ab2} + IR = 0$$

$$-V_{ab2}(1+k) + IR = 0$$

$$V_{ab2} = \frac{IR}{1+k}$$

Therefore from superposition

$$V_{ab} = V_{ab1} + V_{ab2}$$

$$V_{ab} = \frac{V}{1+k} + \frac{IR}{1+k}$$

$$\boxed{V_{ab} = \frac{V+IR}{1+k}\,\mathrm{V}}$$